

DIAEDDIN RIMAWI

Assistant Professor · Software Engineering ·
AI · Cybersecurity
ACADEMIC CV

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PROFILE

Assistant Professor (RTDa) at the Faculty of Engineering, Free University of Bozen-Bolzano. Research spans multimodal procedural activity understanding, data-driven learning and symbolic process mining, green resilience in cyber-physical systems, and AI-enabled cybersecurity. Experienced across university teaching, applied research infrastructure, software engineering, and technical leadership.

CURRENT APPOINTMENT Assistant Professor (RTDa), Free University of Bozen-Bolzano	RESEARCH FOCUS Software engineering, cybersecurity, resilient AI systems, data-driven learning, process mining
HIGHEST DEGREE PhD in Advanced-Systems Engineering, May 2025	TEACHING LEVELS Bachelor, master, doctoral, and professional training

WORK EXPERIENCE

Nov 2025–Present	Assistant Professor (RTDa) <i>Free University of Bozen-Bolzano · Faculty of Engineering · Bolzano, Italy</i> - Conduct research in the FSE+-funded D2PAM project on data-driven procedural activity mining from multimodal streams, working with Oswald Lanz and Marco Montali. - Teach cybersecurity, software and systems security, software quality and metrics, and computer vision across doctoral, master's, and bachelor's programmes. - Supervise student learning and connect research in multimodal learning, process mining, software engineering, and security.
Jan–Nov 2025	Cybersecurity Technologist <i>Cybersecurity Laboratory (CSLab), Free University of Bozen-Bolzano · Bolzano, Italy</i> - Managed core laboratory platforms, including the Cyber Range, and threat-intelligence tooling. - Contributed to research and funding proposals, advised on cybersecurity technologies, and supported applied research in AI-driven security and OT/IT integration. - Prepared and delivered Software and Systems Security laboratory activities and mentored master's students.
Jan 2022–May 2025	Doctoral Researcher <i>Free University of Bozen-Bolzano · Bolzano, Italy</i> - Introduced Green Resilience for collaborative AI systems and developed optimization, game-theoretic, and reinforcement-learning policies that balance recovery performance with environmental impact. - Evaluated decision-making approaches in simulations and robotic cyber-physical systems; published results in journals and international conferences.
Jun–Dec 2023	Research Intern <i>Fraunhofer Italia Research · Bolzano, Italy</i> - Re-engineered a robotic collaborative AI system from bare-metal infrastructure to a Docker-based environment. - Led experiments evaluating automated decision-making frameworks for resilience and sustainability trade-offs.
Sep 2020–Aug 2022	Part-time University Instructor <i>Birzeit University · Birzeit, Palestine</i> - Taught Discrete Mathematics, Database Management Systems, and Linux Operating Systems Laboratory to computer-science and engineering students. - Supervised graduation projects and developed course materials, exercises, assignments, and assessments.
Nov 2020–Dec 2021	Head of Instructors <i>AXSOS Academy</i> - Led instructors and teaching assistants in a full-stack development bootcamp and coordinated curriculum delivery. - Taught Python (Flask/Django), Java (Spring Boot), and JavaScript (MERN) tracks; led development of the academy portal.

Jul 2017–Nov 2020	Software Engineer <i>ASAL Technologies · NVIDIA/Mellanox Firmware Tools team</i> - Developed and verified software in Python, C, C++, Qt, shell scripting, and the FrameMaker Development Kit. - Planned test cases, automated regression testing, developed features, fixed defects, and maintained open-source MSTFlint components.
Sep 2014–Aug 2017	Teaching and Research Assistant <i>Birzeit University · Birzeit, Palestine</i> - Supported computer-science laboratories, prepared projects and course materials, wrote quizzes, graded assignments, and led problem-solving sessions. - Contributed to research on uncertainty in adaptive systems using goal models and genetic-algorithm optimization, demonstrated through a smart-parking case study.

EDUCATION

2022–2025	PhD in Advanced-Systems Engineering <i>Free University of Bozen-Bolzano · Bolzano, Italy</i> - Dissertation topic: Green Resilience of Smart Cyber-Physical Systems. - Supervisors: Barbara Russo and Antonio Liotta. Funded through a PON scholarship for innovation and green studies.
2015–2020	MSc in Software Engineering <i>Birzeit University · Birzeit, Palestine</i> - Research on static and model-based detection of design-pattern usage in Android applications. - Software Engineering Summer School, University of Koblenz-Landau, Germany, July 2016.
2010–2014	Bachelor in Computer Science <i>Birzeit University · Birzeit, Palestine</i> President of the Computer Science Students Club; member of the College Council.
1998–2010	Secondary School of Science <i>Basheer Al-Bargouthy Secondary School · Palestine</i>

TEACHING EXPERIENCE

Current university teaching is presented from doctoral to master's and bachelor's level.

DOCTORAL	Cybersecurity Free University of Bozen-Bolzano · PhD Programme in Advanced-Systems Engineering · EN
MASTER	Software and Systems Security Free University of Bozen-Bolzano · Master in Software Engineering · EN
MASTER	Software Quality and Metrics Free University of Bozen-Bolzano · Master in Software Engineering · EN
BACHELOR	Introduction to Computer Vision Free University of Bozen-Bolzano · Bachelor in Computer Science · EN Free University of Bozen-Bolzano · Bachelor in Electronics and Cyber-Physical Systems Engineering · EN
BACHELOR	Discrete Mathematics Birzeit University · Computer science and engineering · EN/AR
BACHELOR	Database Management Systems Birzeit University · Computer science and engineering · EN/AR
BACHELOR	Linux Operating Systems Laboratory Birzeit University · Computer science · EN/AR
PROFESSIONAL	Computer Vision and Neural Networks Gaza Sky Geeks · Technical training · 2026
PROFESSIONAL	Programming Fundamentals and Object-Oriented Programming in Python Gaza Sky Geeks · Technical training · 2025
PROFESSIONAL	Data Structures and Algorithms Gaza Sky Geeks · Technical training · 2025
BOOTCAMP	Full-stack development tracks AXSOS Academy · Python/Flask/Django; Java/Spring Boot; MERN · 2020–2021

RESEARCH PROJECTS AND FUNDING

2025–Present	D2PAM — Data-driven Procedural Activity Mining from Multi-modal Streams <i>Free University of Bozen-Bolzano · FSE+ funded</i> Research on combining data-driven learning with symbolic process mining for procedural activity understanding from multimodal streams.
2025	Cyber Range acquisition and design <i>Cybersecurity Laboratory, Free University of Bozen-Bolzano</i> Led technical design and procurement contributions for a modular, scalable platform aligned with research and teaching requirements; supported evaluation of European tender submissions.
2025	AI-based security for OT networks (Zero Trust CPS) <i>European Digital Innovation Hub proposal</i> Co-authored the proposal and defined milestones for AI-based intrusion detection and Zero Trust principles in industrial operational-technology networks.
2025	NIS2 compliance and threat modeling for critical infrastructure <i>Public-sector applied-research collaboration</i> Supported NIS2 alignment, threat modeling, attack-surface analysis, and cybersecurity recommendations for a critical-infrastructure provider.
2022–2024	PON Scholarship for Innovation and Green Studies <i>Italian Ministry of University and Research (MUR)</i> Doctoral scholarship awarded.

SKILLS

RESEARCH AND AI Machine learning, deep learning, reinforcement learning, multi-objective optimization, game theory, NLP and sentence transformers, computer vision, process mining	CYBERSECURITY Threat modeling, NIS2 Directive, Zero Trust, OT/IT security, Cyber Range environments, microservice attack detection and performance monitoring
PROGRAMMING Python, C, C++, Java, JavaScript/Node.js, SQL, shell scripting	PLATFORMS AND TOOLS Linux, Git, Docker and containerization, ROS, virtualization, automated testing, Flask, Django, Spring Boot, MERN
SOFTWARE ENGINEERING Software architecture, design patterns, static analysis, software quality, requirements engineering, testing and verification	TEACHING AND LEADERSHIP Curriculum design, lecturer, laboratory teaching, student supervision, project and proposal development

LANGUAGE COMPETENCE

Language	Listening	Reading	Spoken interaction	Spoken production	Writing
Arabic (native)	C2	C2	C2	C2	C2
English	C1	C1	C1	C1	C1
Italian	A1.2	A1.2	A1.2	A1.2	A1.2

Levels follow the Common European Framework of Reference for Languages (CEFR).

CERTIFICATES

- English Language Certificate — Level C1, Free University of Bozen-Bolzano, 2024. [Credential](#)
- Privacy in Research — GDPR Training, Free University of Bozen-Bolzano, 2024. [Credential](#)

PUBLICATIONS

Persistent profiles: [Google Scholar](#) · [ORCID](#) · [DBLP](#) · [Scopus](#).

Journal articles

J1

R. Othman, D. Rimawi, B. Rossi, and B. Russo. **From Attack Descriptions to Vulnerabilities: A Sentence Transformer-Based Approach**. Journal of Systems and Software, 231, 112615, 2026.

J2	D. Rimawi, A. Liotta, M. Todescato, and B. Russo. GResilience: Decision-Making Policies for Trading Off Greenness and Resilience in Online Collaborative AI Systems . IEEE Access, 13, 140867–140888, 2025.
J3	D. Rimawi and S. Zein. A Static Analysis of Android Source Code for Design Patterns Usage . International Journal of Advanced Trends in Computer Science and Engineering, 9(2), 2178–2186, 2020.
Conference and workshop papers	
C1	A. Janes, D. Rimawi, R. T. Othman, A. Avritzer, R. Mirandola, and B. Russo. Toward Performance-Based Detection of Attacks in Microservice Systems . Companion of the 17th ACM/SPEC International Conference on Performance Engineering, 100–101, 2026.
C2	R. Othman, D. Rimawi, B. Rossi, and B. Russo. Predicting Known Vulnerabilities from Attack News: A Transformer-Based Approach . 2nd Joint Italian National Conference on Cybersecurity (ITASEC26), 2026.
C3	D. Rimawi, A. Liotta, M. Todescato, and B. Russo. Modeling Resilience of Collaborative AI Systems . IEEE/ACM 3rd International Conference on AI Engineering (CAIN), 24–29, 2024.
C4	D. Rimawi, A. Liotta, M. Todescato, and B. Russo. CAIS-DMA: A Decision-Making Assistant for Collaborative AI Systems . Product-Focused Software Process Improvement (PROFES), 183–199, 2023.
C5	D. Rimawi, A. Liotta, M. Todescato, and B. Russo. GResilience: Trading Off Between the Greenness and the Resilience of Collaborative AI Systems . IFIP International Conference on Testing Software and Systems (ICTSS), 266–273, 2023.
C6	D. Rimawi. Green Resilience of Cyber-Physical Systems . IEEE International Symposium on Software Reliability Engineering Workshops (ISSREW), 105–109, 2022.
C7	D. Rimawi and S. Zein. A Model-Based Approach for Android Design Patterns Detection . 3rd International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT), 1–10, 2019.

ACADEMIC SERVICE

Journal reviewing

- Reviewer, IEEE Transactions on Automation Science and Engineering (T-ASE), 2026.
- Reviewer, IEEE Access, 2026.
- Reviewer, Information and Software Technology, 2024–2025.

Programme committees and conference reviewing

- Program Committee Member, Posters Track, 6th International Conference on AI Engineering — Software Engineering for AI (CAIN), 2027.
- Program Committee Member, Open Science Track, European Conference on Software Architecture (ECSA), 2026.
- Program Committee Member, 4th International Workshop on Mining Software Repositories Applications for Privacy and Security (MSR4P&S), 2026.
- Program Committee Member, Posters Track, 5th International Conference on AI Engineering — Software Engineering for AI (CAIN), 2026.
- Program Committee Member, Posters & Demos, 16th International Conference on Software Business (ICSOB), 2025.
- Sub-reviewer, Euromicro Conference on Software Engineering and Advanced Applications (SEAA), 2025.